

OptiCrop: Proposed Solution

Solution Summary

OptiCrop is a crop recommendation system that takes soil nutrient levels (N, P, K), temperature, humidity, pH, and rainfall as input, and outputs the most suitable crop using a trained Logistic Regression classifier.

Key Components

- Data pipeline: cleaning, outlier handling, and feature scaling of the crop dataset.
- Model: Logistic Regression classifier over 22 crop classes.
- Interpretability layer: rule-based seasonal grouping (Kharif/Rabi/Summer) and K-Means clustering of soil/climate profiles.
- Delivery: a Flask web application with a simple input form and result page.

Novelty / Value-add

Beyond a raw prediction, OptiCrop groups the same data by season and by K-Means-derived soil profile, giving the user two additional, interpretable views into why a recommendation makes sense.